

Charge is never constant on a nontrivial fibre: the monomial locus of the Kostka–Foulkes polynomials

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Abstract

Question Q380 asked for the pairs (λ, μ) with $\#\text{SSYT}(\lambda, \mu) \geq 2$ for which the Kostka–Foulkes polynomial $K_{\lambda\mu}(t)$ is a monomial — equivalently, the pairs on which the Lascoux–Schützenberger charge statistic is constant on a fibre with something to be constant *on*. **The answer is that there are none**, and the reason is one line: $K_{\lambda\mu}(t)$ is *monic*, so a monomial is forced to be $t^{n(\mu)-n(\lambda)}$ with coefficient 1, and its value at $t = 1$ — the Kostka number — is therefore 1. Hence

$$K_{\lambda\mu}(t) \text{ is a monomial} \iff \#\text{SSYT}(\lambda, \mu) = 1 \iff K_{\lambda\mu}(t) = t^{n(\mu)-n(\lambda)}.$$

The monomial locus coincides exactly with the Kostka-number-one locus, so Q380’s “boundary” is the boundary of $\{K_{\lambda\mu} = 1\}$, and §4 measures it. Three things are proved there: (i) for fixed λ the locus is an *order filter* in the dominance order, hence determined by its minimal elements, of which there are **at most two** for every λ with $|\lambda| \leq 10$; (ii) two exact reductions R_1, R_2 account for all but 10 of the 1044 irreducible dominance pairs with $|\lambda| \leq 10$; (iii) no criterion depending only on the difference vector $\lambda - \mu$, and none depending only on the dominance-slack vector, can decide the question — both are refuted by explicit witness pairs. A closed-form characterisation of $\{K_{\lambda\mu} = 1\}$ is **not** obtained and the obstruction is stated precisely in §5.

Everything is verified by two mechanism-disjoint engines and against Macdonald’s printed tables, on an exhaustive census of all 1719 nonempty dominance pairs with $|\lambda| \leq 10$, of which 1245 are non-vacuous ($\#\text{SSYT} \geq 2$). The consequence for the migration no-go of 5 October, which is what Q380 was gating, is Corollary 7: *the no-go’s conclusion has no non-vacuous instance for charge at all*.

Attribution, stated first. Theorem 3 is a two-line deduction from Macdonald III (6.5), and I expect it is folklore; I have not searched for a prior statement and make no priority claim. Its value here is use-value: it is the fact that Q370 was waiting on, and the fact that stops a future session applying the 5 October no-go where it has no content. The proof of monicity reproduced in Lemma 2 is Macdonald’s, from III §6 Example 4; it is reproduced rather than cited because the whole paper rests on it. What is new here, as far as I can tell, is only the boundary measurement of §4 and the two obstruction statements of §4.4.

1 Notation and the two inputs

Partitions λ, μ of N ; λ' is the conjugate; $\ell(\lambda)$ the number of parts; $n(\lambda) = \sum_{i \geq 1} (i-1)\lambda_i$ (Macdonald I (1.5)). Dominance: $\lambda \supseteq \mu$ iff $\sum_{i \leq k} \lambda_i \geq \sum_{i \leq k} \mu_i$ for all k . We write $\text{SSYT}(\lambda, \mu)$ for the set

of semistandard Young tableaux of shape λ and content μ , and $K_{\lambda\mu} = \#\text{SSYT}(\lambda, \mu)$ for the Kostka number. The Kostka–Foulkes polynomials $K_{\lambda\mu}(t)$ are the entries of the transition matrix $K(t) = M(s, P)$ from Schur to Hall–Littlewood functions, $s_\lambda = \sum_\mu K_{\lambda\mu}(t) P_\mu(x; t)$.

Two facts are taken from Macdonald, *Symmetric Functions and Hall Polynomials*, 2nd ed., III §6 (pp. 238–245), deep-read 9 October 2026, with pp. 239–240 read off page images rendered at 200 dpi rather than off the text layer.

Lemma 1 (Macdonald III (6.2)). *$K(1)$ is the Kostka matrix: $K_{\lambda\mu}(1) = K_{\lambda\mu} = \#\text{SSYT}(\lambda, \mu)$. Also $K_{\lambda\mu}(t) = 0$ unless $\lambda \supseteq \mu$, and $K_{\lambda\lambda}(t) = 1$.*

Lemma 2 (Macdonald III (6.5)(ii)). *If $\lambda \supseteq \mu$ then $K_{\lambda\mu}(t)$ is monic of degree $n(\mu) - n(\lambda)$.*

Proof (Macdonald III §6, Example 4), reproduced. Let $n \geq \ell(\mu)$, let $\varepsilon_1, \dots, \varepsilon_n$ be the standard basis of $\mathbb{Z}^n$, $R^+ = \{\varepsilon_i - \varepsilon_j : i < j\}$, and $\delta = (n-1, n-2, \dots, 1, 0)$. For $\xi \in \mathbb{Z}^n$ with $\sum \xi_i = 0$ set

$$P(\xi; t) = \sum_{(m_\alpha)} t^{\sum_\alpha m_\alpha},$$

summed over families $(m_\alpha)_{\alpha \in R^+}$ of non-negative integers with $\xi = \sum_\alpha m_\alpha \alpha$; this is the Lusztig t -analogue of Kostant’s partition function. Writing $\xi = \sum_{i < n} \eta_i(\varepsilon_i - \varepsilon_{i+1})$ with $\eta_i = \xi_1 + \dots + \xi_i$, one has $P(\xi; t) \neq 0$ iff all $\eta_i \geq 0$, and then $P(\xi; t)$ is monic of degree $\sum_i \eta_i = \langle \xi, \delta \rangle$. (Monic: the unique family of maximal total weight is $m_{\varepsilon_i - \varepsilon_{i+1}} = \eta_i$ and all other $m_\alpha = 0$, because replacing a simple-root unit by a longer root strictly lowers $\sum m_\alpha$.)

By III (6.3), $K(t)'$ is the matrix of $\prod_{i < j} (1 - tR_{ij})^{-1}$, so $K_{\lambda\mu}(t)$ is the coefficient of $x^{\lambda+\delta}$ in $\sum_{w \in S_n} \varepsilon(w) w \left(x^{\mu+\delta} \prod_{i < j} (1 - tx_i^{-1}x_j)^{-1} \right)$, whence

$$K_{\lambda\mu}(t) = \sum_{w \in S_n} \varepsilon(w) P(w^{-1}(\lambda + \delta) - (\mu + \delta); t). \quad (1)$$

For the degree of the w -term,

$$\langle w^{-1}(\lambda + \delta) - (\mu + \delta), \delta \rangle = \langle \lambda + \delta, w\delta \rangle - \langle \mu + \delta, \delta \rangle \leq \langle \lambda + \delta, \delta \rangle - \langle \mu + \delta, \delta \rangle = n(\mu) - n(\lambda),$$

using $\sum_i (n-i)\lambda_i = (n-1)|\lambda| - n(\lambda)$. The inequality is the rearrangement inequality applied to the two strictly decreasing vectors $\lambda + \delta$ and δ , and it is *strict* for every $w \neq 1$. The $w = 1$ term is $P(\lambda - \mu; t)$, which is nonzero precisely because $\lambda \supseteq \mu$, and is monic of degree $n(\mu) - n(\lambda)$. Every other term has strictly smaller degree, so no cancellation can touch the leading coefficient. $\square$

We will also use, but only in Corollary 5, the Lascoux–Schützenberger theorem, Macdonald III (6.5)(i): $K_{\lambda\mu}(t) = \sum_{T \in \text{SSYT}(\lambda, \mu)} t^{c(T)}$, c the charge of the reading word. **The main theorem does not depend on it.**

2 The theorem

Theorem 3. *Let $\lambda \supseteq \mu$ be partitions of N . The following are equivalent.*

- (i) $K_{\lambda\mu}(t)$ has exactly one nonzero coefficient.
- (ii) $K_{\lambda\mu}(t) = t^{n(\mu) - n(\lambda)}$.

(iii) $\#SSYT(\lambda, \mu) = 1$.

Proof. (i) $\Rightarrow$ (ii). Write $K_{\lambda\mu}(t) = ct^d$ with $c \neq 0$. By Lemma 2 the leading coefficient is 1 and the degree is $n(\mu) - n(\lambda)$; a single-term polynomial has its only coefficient as its leading coefficient, so $c = 1$ and $d = n(\mu) - n(\lambda)$.

(ii) $\Rightarrow$ (iii). Evaluate at $t = 1$ and apply Lemma 1: $\#SSYT(\lambda, \mu) = K_{\lambda\mu}(1) = 1$.

(iii) $\Rightarrow$ (i). $K_{\lambda\mu}(t)$ has non-negative integer coefficients (Lemma 1 with III (6.5)(i), or Lusztig's geometric interpretation, III §6 Example 6) summing to $K_{\lambda\mu}(1) = 1$, so exactly one coefficient is nonzero. $\square$

Corollary 4 (the answer to Q380). $\{(\lambda, \mu) : \#SSYT(\lambda, \mu) \geq 2 \text{ and } K_{\lambda\mu}(t) \text{ is a monomial}\} = \emptyset$.

Corollary 5 (charge separates). *If $\#SSYT(\lambda, \mu) \geq 2$ then the charge statistic takes at least two distinct values on $SSYT(\lambda, \mu)$. More precisely, $\max_T c(T) = n(\mu) - n(\lambda)$ and it is attained by exactly one tableau.*

Proof. By III (6.5)(i) the multiset of charges is the coefficient list of $K_{\lambda\mu}(t)$; by Lemma 2 the top coefficient is 1 and the top degree is $n(\mu) - n(\lambda)$, so the maximum charge is attained once. If $\#SSYT(\lambda, \mu) \geq 2$, some other tableau has strictly smaller charge. $\square$

Remark 6. Corollary 5 is the statement with content. Theorem 3 says the *only* way for charge to be constant on a fibre is for the fibre to have nothing to be constant on — which is exactly the degenerate situation that Q380 was constructed to exclude. Equivalently: *the unique maximal-charge tableau is the entire obstruction*, and it exists in every fibre.

2.1 What this does to the migration no-go, and to Q370

Corollary 7 (the no-go is vacuous for charge). *Let ϕ be any statistic on $SSYT(\lambda, \mu)$ and suppose the no-go of 2026-10-05-c3-migration-length-grading.tex applies to it — i.e. ϕ is a potential difference for some monotone process on the fibre, so that the fibre generating function $\sum_T q^{\phi(T)}$ is a monomial cq^{ℓ_0} . Taking $\phi = c$, that conclusion holds if and only if $c = 1$. Therefore the no-go, applied to the charge statistic on Kostka–Foulkes fibres, has **no non-vacuous instance whatsoever**: its hypothesis forces precisely the fibres on which its conclusion is empty.*

Proof. Immediate from Theorem 3: “ $\sum_T q^{c(T)}$ is a monomial” is equivalent to $\#SSYT(\lambda, \mu) = 1$, which is $c = 1$. $\square$

This settles the gating question behind Q370 (*why does charge evade the potential no-go: by reversibility, by branching, or because $K_{\lambda\mu}(t)$ is not a fibre generating function of a monotone process at all?*) in the following sense, and only in this sense: **charge does not need to evade the no-go, because the no-go has no purchase on it**. Whenever the fibre has ≥ 2 elements, Corollary 5 says $\sum_T q^{c(T)}$ is not a monomial, so by the contrapositive of the no-go, *charge is not a potential difference for any monotone process on that fibre*. The third horn of Q370 is the right one, and it is now a theorem rather than an assertion. The amendment owed to the 5 October artifact (its [A3]) is therefore: on the Kostka–Foulkes side the $c = 1$ degeneracy is not an accident of the sample, it is forced.

3 Engines, controls, and the census

Engine A (charge). $\text{SSYT}(\lambda, \mu)$ built by Gelfand–Tsetlin horizontal-strip growth; each tableau then verified *entrywise* for shape, content, weak rows and strict columns; $K_{\lambda\mu}(t) = \sum_T t^{c(T)}$ with charge computed by the Lascoux–Schützenberger standard-subword rule.

Engine B (Weyl alternating sum). Equation (1): a signed sum over S_n of values of the t -analogue of Kostant’s partition function, the latter computed by exhaustive enumeration of non-negative integer families $(m_\alpha)_{\alpha \in R^+}$. No tableau, no reading word and no charge occurs anywhere in Engine B. The two engines therefore differ in mechanism, not merely in basis.

Engine C (published). Macdonald’s printed $K(t)$ matrices for $n \leq 5$ (III §6, p. 239), read off a 200 dpi page image.

check	scope	result
A vs. B	all 53 dominance pairs, $N \leq 5$	0 mismatches
A vs. C	all printed entries, $n \leq 5$	0 mismatches

Remark 8 (the text layer lies). The brief’s warning was correct and it fired. The OCR text layer of p. 239 column-shifts the $n = 5$ matrix, yielding $K_{(2,2,1),(2,1^3)}(t) = t$ — a plausible-looking monomial on a fibre that in fact has two tableaux and $K = t + t^2$. Had that reading been trusted, this paper would have opened with a counterexample to its own theorem. It was caught by hand before it was caught by the engines. Pages that carry load are read as images.

Planted controls. Before reading any verdict:

- *The statistic varies with the claim.* The detector is $\#\{\text{nonzero coefficients}\}$. On $(\lambda, \mu) = ((3, 1), (1^4))$ it reads 3. Replacing c by a constant function on that fibre makes it read 1 with $\#\text{SSYT} = 3$, i.e. the planted pair *does* enter the target set of Corollary 4. The detector is not a constant function of the thing it tests, and the empty target set below is a measurement, not a tautology.
- *The entrywise tableau check fires.* Corrupting row order, a column inequality, or the content of a genuine tableau is rejected in all three cases.
- *Negative control.* All 201 pairs with $N \leq 7$ failing dominance return $\text{SSYT} = \emptyset$ (not a crash, not garbage).
- *Cardinality is not content.* All 2767 tableaux generated for $N \leq 8$ pass the entrywise check, and no generated set contains a duplicate.

The census. All pairs of partitions of N , $1 \leq N \leq 10$.

N	#partitions	pairs	SSYT $\neq \emptyset$	#SSYT ≥ 2	K monomial
1	1	1	1	0	1
2	2	4	3	0	3
3	3	9	6	1	5
4	5	25	15	4	11
5	7	49	28	12	16
6	11	121	64	32	32
7	15	225	116	72	44
8	22	484	238	161	77
9	30	900	430	320	110
10	42	1764	818	643	175
total		3582	1719	1245	474

The non-vacuous population is 1245, so the range is not vacuous; it first becomes nonempty at $N = 3$. Over all 1719 nonempty pairs:

$$\begin{aligned} \#\{\deg K_{\lambda\mu} \neq n(\mu) - n(\lambda)\} &= 0, & \#\{\text{leading coefficient} \neq 1\} &= 0, \\ \#\{\#\text{SSYT} \geq 2 \text{ and } K_{\lambda\mu}(t) \text{ a monomial}\} &= 0, \end{aligned}$$

and $474 = 1719 - 1245$ monomial pairs, i.e. the monomial count equals the $\#\text{SSYT} = 1$ count at every N separately. This is Theorem 3 and Lemma 2, measured.

The two-row family (the brief's vacuity check). For all $N \leq 12$ and all $j \leq k$, $K_{(N-j,j),(N-k,k)}(t) = t^{k-j}$ with $\#\text{SSYT} = 1$: 138 pairs, 0 failures. So `lem:mono of 2026-10-07-two-part-green-polynomials.tex` is indeed a *singleton* fibre and is vacuous for Q380. This is now also a corollary of Proposition 12 below rather than a computation.

4 The boundary

By Theorem 3 the monomial locus *is* the Kostka-number-one locus

$$\mathcal{M}(\lambda) = \{\mu : K_{\lambda\mu} = 1\} \subseteq \{\mu : \lambda \triangleright \mu\},$$

so Q380's deliverable — the boundary — is the boundary of $\mathcal{M}(\lambda)$.

4.1 $\mathcal{M}(\lambda)$ is an order filter

Proposition 9. *If $\lambda \triangleright \nu \triangleright \mu$ then $K_{\lambda\nu} \leq K_{\lambda\mu}$. Consequently $\mathcal{M}(\lambda)$ is an order filter (up-set) in the dominance interval $[(1^N), \lambda]$, and is determined by its set of minimal elements.*

Proof sketch, and the honest status. This monotonicity of Kostka numbers in the dominance order on the content is classical. Since $K_{\lambda\mu}$ depends only on the multiset of parts of μ , it suffices to treat $\nu = (\dots, a+1, b-1, \dots)$, $\mu = (\dots, a, b, \dots)$ with $a \geq b$ in two adjacent positions $i, i+1$. Restricting the crystal $B(\lambda)$ to the i -th $\mathfrak{sl}_2$ decomposes $\bigsqcup_c \text{SSYT}(\lambda, \cdot)$ into i -strings, along each of which the multiset of values of $m_i - m_{i+1}$ is symmetric about 0 and consists of consecutive integers of one parity; hence $\#\{m_i - m_{i+1} = d\}$ is weakly decreasing in $|d|$, which is the claim. **I have not written this argument out in full** — the string decomposition is standard but I am citing it, not proving it here. Verified computationally on all 6750 triples $\lambda \triangleright \nu \triangleright \mu$ with $N \leq 9$: 0 violations. $\square$

Theorem 10 (the measured boundary). *For every partition λ with $|\lambda| \leq 10$, the filter $\mathcal{M}(\lambda)$ has at most two minimal elements.*

This is a measurement over all $42 + 30 + \dots = 136$ partitions with $|\lambda| \leq 10$, not a proof, and I do not conjecture that 2 is a bound for all N — see §5. Examples at $N = 10$ with exactly two minimal elements:

$$\begin{array}{ll}
(8, 2) & \rightarrow \{(8, 1, 1), (5, 5)\} \\
(7, 3) & \rightarrow \{(7, 1^3), (5, 5)\} \\
(6, 4) & \rightarrow \{(6, 1^4), (5, 5)\} \\
(6, 3, 1) & \rightarrow \{(6, 2, 2), (5, 4, 1)\} \\
(5, 3, 1, 1) & \rightarrow \{(5, 2, 2, 1), (4, 4, 1, 1)\} \\
(6, 2, 2) & \rightarrow \{(4, 4, 1, 1), (4, 3, 3)\} \\
(5, 5) & \rightarrow \{(5, 1^5), (4, 3, 3)\} \\
(4, 4, 2) & \rightarrow \{(4, 4, 1, 1), (4, 3, 3)\} \\
(4, 4, 1, 1) & \rightarrow \{(4, 2, 2, 2), (3, 3, 3, 1)\} \\
(4, 3, 3) & \rightarrow \{(4, 3, 1^3), (4, 2, 2, 2)\}
\end{array}$$

4.2 Two exact reductions

Proposition 11 (R_1 , first row). *If $\mu_1 = \lambda_1$ then $K_{\lambda\mu} = K_{(\lambda_2, \lambda_3, \dots), (\mu_2, \mu_3, \dots)}$.*

Proof. In $T \in \text{SSYT}(\lambda, \mu)$ every 1 lies in row 1 (a 1 in row $i \geq 2$ would need a smaller entry above it) and, rows being weakly increasing, the 1s occupy the leftmost cells of row 1. As $\mu_1 = \lambda_1$ they fill row 1 exactly. Deleting row 1 is injective into $\text{SSYT}((\lambda_2, \dots), (\mu_2, \dots))$; conversely, prepending a row of λ_1 ones to any such tableau is column-strict because all its entries are ≥ 2 . $\square$

Proposition 12 (R_2 , first column). *If $\ell(\mu) = \ell(\lambda) = m$ then $K_{\lambda\mu} = K_{\hat{\lambda}\hat{\mu}}$ with $\hat{\lambda} = (\lambda_1 - 1, \dots, \lambda_m - 1)$, $\hat{\mu} = (\mu_1 - 1, \dots, \mu_m - 1)$ (zero parts dropped).*

Proof. Entries of T lie in $\{1, \dots, \ell(\mu)\} = \{1, \dots, m\}$, and column 1 has m cells with strictly increasing entries, so column 1 is $(1, 2, \dots, m)^T$. Deleting it gives a semistandard tableau of shape $\hat{\lambda}$ and content $\hat{\mu}$. Conversely, given such an S , prepend the column $(1, \dots, m)^T$: columns stay strict, and rows stay weakly increasing because in any column-strict tableau with positive entries $S(i, 1) \geq i$. The two maps are mutually inverse. $\square$

Note $\lambda \succeq \mu$ forces $\ell(\mu) \geq \ell(\lambda)$ and $\mu_1 \leq \lambda_1$, so R_1 and R_2 apply at the two opposite extremes. Both verified on all 204 pairs with $N \leq 9$ to which they apply: 0 failures.

Corollary 13. *Each of the following has $K_{\lambda\mu} = 1$, hence $K_{\lambda\mu}(t)$ a monomial.*

- (a) $\lambda = (N)$, any μ (Macdonald III §6, Example 1: $K_{(N)\mu}(t) = t^{n(\mu)}$).
- (b) $\lambda = (N - j, j)$ and $\mu = (N - k, k)$ with $j \leq k$: apply R_2 j times to reach a one-row shape. So the two-row/two-row slice is a singleton fibre throughout.
- (c) Adjacent transfer: $\mu = \lambda - r\varepsilon_k + r\varepsilon_{k+1}$ for some $k \geq 1$, $r \geq 1$, whenever this is a partition.

Proof of (c). By R_1 applied $k - 1$ times we may take $k = 1$, so $\mu = (\lambda_1 - r, \lambda_2 + r, \lambda_3, \dots)$. In $T \in \text{SSYT}(\lambda, \mu)$ the $\lambda_1 - r$ ones fill the left of row 1. Entries in rows ≥ 3 are ≥ 3 , so all $\mu_2 = \lambda_2 + r$ twos lie in rows 1 and 2; row 2 holds at most λ_2 of them and row 1 has exactly r free cells, so row 1 = $1^{\lambda_1 - r} 2^r$ and row 2 = 2^{λ_2} , forced. (This is column-strict because $\mu_1 \geq \mu_2$ gives $\lambda_1 - r \geq \lambda_2 + r \geq \lambda_2$.) The remaining letters are $3^{\lambda_3}, 4^{\lambda_4}, \dots$ on the shape $(\lambda_3, \lambda_4, \dots)$, where the only semistandard filling is the one with row i constant — i.e. $K_{\nu\nu} = 1$. $\square$

Verified: all 250 adjacent-transfer pairs with $N \leq 11$ have $K = 1$, and all 250 are in fact killed by R_1 and R_2 alone.

4.3 The irreducible core

Call (λ, μ) *irreducible* if $\ell(\lambda) \geq 2$, $\mu_1 < \lambda_1$ and $\ell(\mu) > \ell(\lambda)$, i.e. neither R_1 nor R_2 applies and λ is not a single row.

Theorem 14 (measured). *Of the 1044 irreducible dominance pairs with $N \leq 10$, exactly 10 have $K_{\lambda\mu} = 1$:*

$$\begin{array}{lll} ((3, 3), (2, 2, 2)) & ((4, 4), (3, 3, 2)) & ((5, 5), (4, 4, 2)) \\ ((5, 5), (4, 3, 3)) & ((4, 2, 2), (3, 3, 1, 1)) & ((5, 2, 2), (4, 3, 1, 1)) \\ ((6, 2, 2), (5, 3, 1, 1)) & ((6, 2, 2), (4, 4, 1, 1)) & ((5, 3, 2), (4, 4, 1, 1)) \\ ((4, 2, 2, 2), (3, 3, 2, 1, 1)) & & \end{array}$$

A third exact symmetry, rectangular complementation — with $r = \max(\ell(\lambda), \ell(\mu))$, $c = \lambda_1$, $K_{\lambda\mu} = K_{\lambda^* \mu^*}$ where $\lambda^* = (c - \lambda_r, \dots, c - \lambda_1)$, $\mu^* = (c - \mu_r, \dots, c - \mu_1)$ (the GL_r duality $V_\lambda^* \otimes \det^c$) — disposes of four of the ten, sending them to one-row shapes:

$$\begin{array}{ll} ((3, 3), (2, 2, 2)) \mapsto ((3), (1^3)), & ((4, 4), (3, 3, 2)) \mapsto ((4), (2, 1, 1)), \\ ((5, 5), (4, 4, 2)) \mapsto ((5), (3, 1, 1)), & ((5, 5), (4, 3, 3)) \mapsto ((5), (2, 2, 1)). \end{array}$$

The remaining six are not reduced by it; three of them — $((4, 2, 2), (3, 3, 1, 1))$, $((5, 3, 2), (4, 4, 1, 1))$, $((4, 2, 2, 2), (3, 3, 2, 1, 1))$ — are fixed points of the complementation. All six have $\lambda - \mu$ a sum of *two disjoint* adjacent transfers, e.g. $\lambda - \mu = (1, -1, 1, -1)$ for the first three; but almost all pairs of that form have $K \geq 2$, so this is a description of the six, not a criterion.

4.4 Two things the boundary is not a function of

Theorem 15. (a) *No criterion depending only on the difference vector $\lambda - \mu$ can decide whether $K_{\lambda\mu} = 1$. Witness: $\lambda - \mu = (1, 1, -2)$ for both $((3, 3), (2, 2, 2))$, where $K = 1$ and $K_{\lambda\mu}(t) = t^3$, and $((4, 3), (3, 2, 2))$, where $K = 2$ and $K_{\lambda\mu}(t) = t^2 + t^3$.*

(b) *No criterion depending only on the dominance-slack vector $s_k = \sum_{i \leq k} (\lambda_i - \mu_i)$ can decide it. Witness: $s = (2, 1, 0)$ for both $((4, 1, 1), (2, 2, 2))$, where $K = 1$ and $K_{\lambda\mu}(t) = t^3$, and $((5, 1), (3, 2, 1))$, where $K = 2$ and $K_{\lambda\mu}(t) = t^2 + t^3$.*

Proof. The four Kostka numbers and four polynomials are computed by both engines and listed above; the two difference vectors (resp. slack vectors) are equal by inspection. $\square$

These two are worth recording because they are exactly the two shapes of answer I would have reached for first. (a) kills “ μ is obtained from λ by the following box moves”; (b) kills “ λ and μ touch in the following positions”. Any correct criterion must read λ and μ jointly, not through either of these projections.

5 Gaps

Gap 16. A closed-form characterisation of $\{(\lambda, \mu) : K_{\lambda\mu} = 1\}$ is **not** obtained. What is obtained is: the filter structure (Prop. 9), the two exact reductions, the proved families of Cor. 13, the measured statement that the reductions leave only 10 of 1044 irreducible pairs for $N \leq 10$, and the proved fact (Thm. 15) that no criterion in $\lambda - \mu$ or in the slack vector can exist. The precise shape of the

obstruction is the six-element post-complementation core: it is nonempty, so “ $K_{\lambda\mu} = 1$ iff (λ, μ) reduces to a one-row shape under R_1, R_2, R_3 ” is *false*.

Gap 17. Theorem 10 (≤ 2 minimal elements) and Theorem 14 (10 of 1044) are measurements for $N \leq 10$, with no proof and no conjecture attached. In particular I do *not* claim the bound 2 persists; the honest reading is that the exceptional set is sparse in this range, and sparse is not empty.

Gap 18. Prop. 9 is cited, not proved: the $\mathfrak{sl}_2$ -string decomposition of $B(\lambda)$ is standard and I have sketched how it gives monotonicity, but I have not written the argument out. It is used only for the *structure* of the boundary (Thm. 10), never for Theorem 3.

What the main theorem does and does not rest on. Theorem 3 rests on Lemmas 1 and 2 only, both from Macdonald III §6, both deep-read, and Lemma 2’s proof is reproduced in full. It does *not* rest on the Lascoux–Schützenberger theorem III (6.5)(i), which enters only in Corollary 5 where the word “charge” is used, and not in the computations, which agree across two mechanism-disjoint engines and with Macdonald’s printed tables.

Numerals appearing in the answer. Per the brief’s §4(f): the only constants in the statements are 1 (the Kostka number, the leading coefficient, and the number of maximal-charge tableaux — these are three different occurrences of 1 and Lemma 2 is exactly the bridge between the second and the third), and the exponent $n(\mu) - n(\lambda)$, which at $\lambda = \mu$ is 0 and must not be confused with the charge of the superstandard tableau: they agree, but for the reason that $K_{\lambda\lambda}(t) = 1$, not by coincidence. The 2 of Theorem 10 is a measured cardinality of a set of minimal elements; it is not ℓ of anything, not a Bell number, and the first place a competing reading would diverge is $N = 11$, which is not computed.

Code. `proofs/code-1009-q380/`: `kf.py` (Engine A), `kostant.py` (Engine B), `census.py`, `controls.py`, `reduce.py`.